`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

TODO: Write a clean 120–150 word abstract after we decide the final empirical scope.

Nina: Should the paper pitch itself primarily as a software paper with one engineering/scientific case study, or as a software paper plus empirical ML paper?

Nina: For the abstract, should we keep the empirical claim to PCam only unless a second experiment is complete enough to defend?

Candidate abstract structure:

- Problem: selective G-bispectra are theoretically attractive but not available as practical ML software.
- Main artifact: a PyTorch library implementing selective or full bispectral invariants across several group/domain pairs.
- Main technical point: implementing inversion for the octahedral group required resolving a concrete algorithmic failure of the bootstrap strategy.
- Empirical point: evaluate the library as an invariant pooling primitive on PatchCamelyon, with final claims deferred until sweeps are complete.
- Scope note: do not mention steerable-to-Fourier in this version.

Nina: We need a 3D dataset to test the Octahedral group (`OctaonOcta`). This is required for the empirical story to cover our main technical contribution.

Nina: Deadline is April 14 for internal review.

Nina: We need to look again at the state of the art for the PCam dataset — make sure our baselines and comparisons are up to date.

Nina: We need to do a Pareto analysis for the amount of data (accuracy vs. training set size across pooling methods).

1 Introduction

Equivariant neural networks enforce symmetry by construction, but they still require an invariant representation at the point where a prediction must no longer depend on orientation. In practice, invariant pooling remains a weak point: standard choices such as norm pooling, gate-based pooling, or max/sum aggregation are usable, but they do not preserve all equivariant information. The bispectrum offers a mathematically principled alternative, since for many groups it yields exact invariants with reconstruction guarantees.

The obstacle is practical rather than conceptual. Existing bispectral constructions are scattered across theory papers, domain-specific derivations, and small implementations, with no general-purpose PyTorch library covering the finite-group and compact-group cases relevant to modern geometric deep learning. Even for groups where selectivity reduces the coefficient count from $O(|G|^2)$ to $O(|G|)$, a user still needs a numerically stable implementation, a consistent API, inversion routines, tests, and benchmarked runtime behavior.

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This paper presents `bispectrum`, a PyTorch library for computing bispectral invariants and, where available, their inverses. The library supports cyclic, dihedral, torus, disk, sphere, and octahedral settings as differentiable `nn.Modules` intended to plug into standard ML workflows. Our emphasis in this draft is on the software design and on one specific engineering contribution that arose during implementation: inversion for the octahedral group.

TODO: Rewrite the introduction once the final empirical scope is fixed.

Decision needed: Keep the paper centered on the library + octahedral inversion, or broaden to "library + empirical study of invariant pooling"?

Nina: Do we want the intro to frame the library as infrastructure for geometric deep learning in general, or specifically as a practical invariant nonlinearity for equivariant CNNs?

Proposed contributions for this version. This scaffold assumes the paper will make the following contributions:

1. **Software contribution.** A PyTorch library implementing bispectral invariants for multiple group/domain pairs with a consistent `nn.Module`-style interface.
2. **Engineering/scientific contribution.** An analysis of why the standard selective-bispectrum bootstrap inversion fails for the octahedral group, together with a working bootstrap + Levenberg–Marquardt correction.
3. **Empirical contribution.** A task-driven evaluation of the library as an invariant pooling primitive, with final claims currently left open pending completed sweeps.

Nina: Is this the right level of ambition for a one-month deadline, or should we cut to only (1) + (2) with experiments framed as validation rather than broad empirical claims?

2 Mathematical Background

2.1 Equivariant features and invariant maps

TODO: Write a short background subsection on equivariant CNNs and invariant pooling. Keep this focused and non-controversial.

Suggested points:

- Equivariant layers transform predictably under a group action.
- Downstream prediction tasks often require a representation invariant to that action.
- Existing invariant maps in equivariant architectures are typically chosen for convenience rather than completeness.
- The paper only needs enough background to motivate the bispectrum as a candidate invariant primitive.

2.2 The G-bispectrum and selective bispectra

TODO: Write a concise subsection giving the bispectrum definition and the idea of selectivity.

Minimal content to include:

- finite-group Fourier transform notation,
- the bispectrum as a triple-product / Clebsch–Gordan construction,
- completeness as orbit separation under appropriate nondegeneracy assumptions,
- selective bispectrum as a reduced coefficient set with the same recovery guarantees in the settings covered by prior theory.

Nina: Should this section stay abstract and general, or should we immediately specialize notation toward the modules in the library table?

3 The `bispectrum` Library

This section should read like a software paper, not a tutorial. The goal is to explain what the library covers, what abstractions it exposes, and what design choices were made to keep the implementation mathematically legible and usable in PyTorch workflows.

3.1 Design principles

TODO: Write this as a design-principles section, borrowing the "manifesto" framing from strong software papers.

Candidate principles:

- **Mathematical naming.** Module names encode both the acting group and the signal domain, e.g. `CnonCn`, `DnonDn`, `SO2onDisk`, `SO3onS2`, `OctaonOcta`.
- **`nn.Module` interface.** Every transform is packaged as a module with a forward pass and, where available, an inversion routine.
- **Raw-signal input.** Users pass signals in their natural domain; the internal Fourier or harmonic transform is handled by the module.
- **Registered buffers for mathematical objects.** Precomputed Clebsch–Gordan matrices, harmonic bases, and related constants move correctly across device and dtype.
- **Selective when justified.** Modules expose selective outputs when the mathematics and implementation support them; otherwise the paper should state the limitation explicitly.

Nina: Do we want to include a short subsection here on what the library deliberately does *not* try to unify, to avoid overclaiming API uniformity?

3.2 Supported modules

TODO: Fill in the table only with claims that the current package actually supports. Avoid wording that implies full-mode support everywhere.

Table 1: Current scope of the `bispectrum` library.

Module	Group / Domain	Output mode	Inversion status
`CnonCn`	C_n on C_n	selective + full	implemented
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full via cyclic case	implemented
`TorusOnTorus`	product cyclic groups on torus	selective + full	implemented
`DnonDn`	D_n on D_n	selective	implemented
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective	implemented
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	full only	not implemented
`OctaonOcta`	octahedral group	selective	implemented

TODO: Verify exact wording for the octahedral domain and whether we want to call it "on $\mathbb{R}^3$", "on the octahedral group", or something more implementation-specific.

3.3 Example API

TODO: Include one short code snippet that actually matches the public package API.

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

Nina: Should the paper include a single minimal code example only, or one example per modality (finite group, disk, sphere)?

3.4 Octahedral inversion as an implementation case study

The clearest technical case study in the library is inversion for the octahedral group. Implementing `OctaonOcta` exposed a failure mode of the standard bootstrap strategy: the selective bispectrum remains complete, but the sequential reconstruction procedure breaks because the ambiguity recovered at the first nontrivial irrep lives in $\mathrm{SO}(3)$, while the octahedral Clebsch–Gordan structure only block-diagonalizes the discrete group action.

TODO: Write this section as a focused engineering/scientific contribution, not as a broad new theory section.

Core points to cover:

- what the bootstrap strategy tries to do,
- why it works in cyclic and dihedral settings,
- what fails in the octahedral case,
- how the library resolves this in practice using bootstrap initialization plus Levenberg–Marquardt refinement and multiple restarts.

Nina: Should this be the central scientific contribution in the paper, with the rest of the library framed as the enabling software context?

4 Benchmarks

TODO: Add benchmark figures only after the benchmark script and naming are cleaned up.

TODO: Decide whether to present both coefficient-count scaling and wall-clock timing, or to prioritize coefficient-count scaling plus one throughput plot.

Nina: Do we want to report only the benchmark regimes that are already stable and reproducible, or keep broader benchmark ambitions in appendix as future work?

Candidate benchmark questions for co-author review:

- Which benchmark best supports the software-paper narrative: coefficient scaling, forward runtime, inversion runtime, memory, or all of the above?
- Should runtime figures compare only supported modes, to avoid paper/code mismatch?
- Do we need two GPU types for credibility, or is one clean benchmarked setup enough for submission?

5 Experiments

5.1 2D histopathology classification under cyclic symmetry

The PatchCamelyon (PCam) benchmark Veeling et al. [2018] consists of 327,680 colour patches (96×96 px) extracted from Camelyon16 whole-slide images of lymph node sections. The task is binary classification: whether the central 32×32 region contains metastatic tissue. Rotation invariance is physically meaningful here, since tissue orientation is arbitrary in histopathology slides and classifiers should be insensitive to it.

Published baselines. Table 2 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark.

Architecture and experimental design. We use a C_8-equivariant DenseNet backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. but replacing the final invariant map. Five model variants differ *only* in their nonlinearity and terminal invariant pooling:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling.

Table 2: Published baselines on PatchCamelyon from Veeling et al. Veeling et al. [2018], trained on 100% data. Our models operate at 68K–920K parameters and are trained on only 10% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

Table 3: PCam results at the $\sim$100K parameter budget (10% training data, seed 42). All equivariant models use the same C_8-DenseNet backbone; only the nonlinearity and invariant map differ.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	0.928	0.853
Gated	136K	0.924	0.856
Bispectrum (`CnonCn`)	80K	0.933	0.880
Fourier-ELU	110K	0.937	0.857
NormReLU	110K	**0.954**	**0.887**

- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via `CnonCn`$(n{=}8)$ as the terminal invariant map.

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from $\sim$30K to $\sim$1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW (lr=10^{-3}, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

Results. Figure 1 presents the Pareto analysis. The key findings are:

Pareto efficiency. At matched parameter budgets ($\sim$100K), equivariant models consistently outperform the augmentation-trained standard CNN (AUC 0.928). NormReLU achieves the highest AUC (0.954) at this operating point, followed by Fourier-ELU (0.937) and bispectrum (0.933). However, scaling behavior differs: as parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all *degrade* (by 1.8, 2.6, and 0.9 AUC points respectively), while bispectrum *improves* (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data efficiency. Panel (c) compares 1% vs. 10% training data at growth rate 12. The bispectrum shows the largest absolute AUC at 1% data (0.875) and the largest gain from 1% to 10% (+6.8 points), consistent with the hypothesis that complete invariants provide a stronger inductive bias when data is scarce.

Comparison with published work. Our best equivariant model (NormReLU, 110K params, AUC 0.954 on 10% data) approaches the published P4M-DenseNet result of Veeling et al. (119K params, AUC 0.963 on 100% data), despite using an order of magnitude less training data. The bispectrum model at 80K parameters (AUC 0.933) exceeds the published DenseNet baseline (128K params, AUC 0.951 on full data) while trained on 10$\times$ fewer examples, demonstrating the sample efficiency of equivariant architectures with principled invariant pooling.

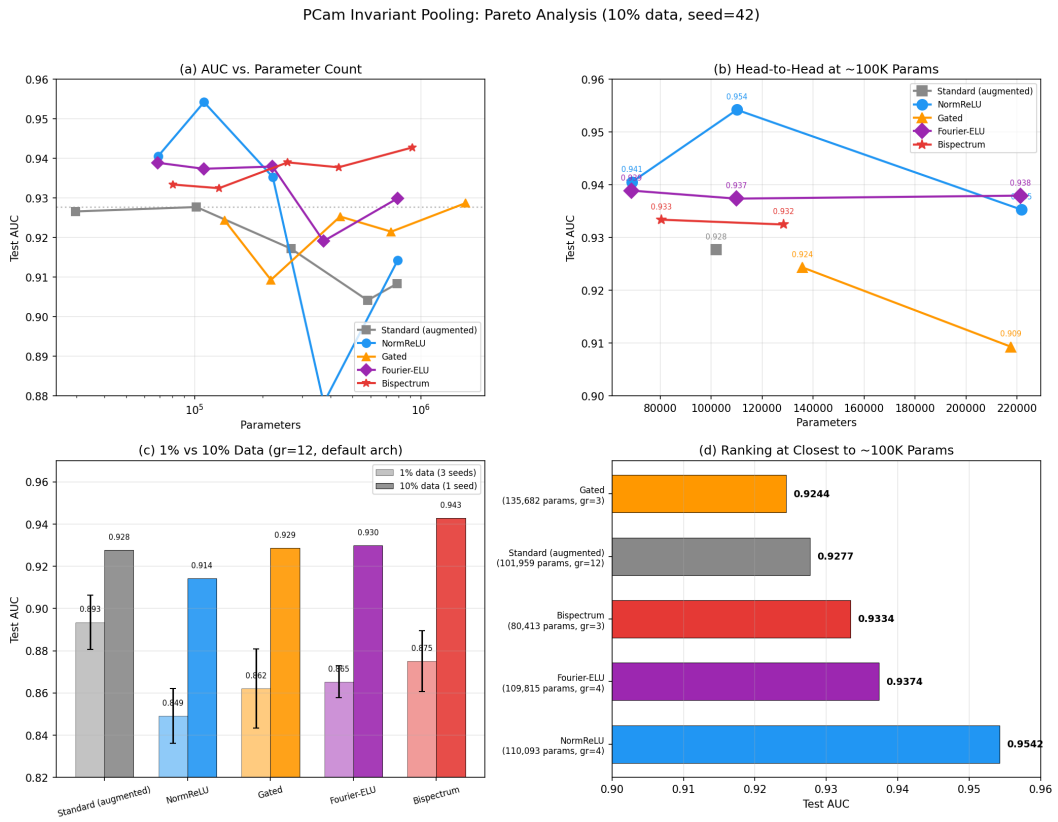

Figure 1: PCam Pareto analysis (10% training data, seed 42). (a) Test AUC vs. parameter count across all five pooling variants. (b) Head-to-head comparison at $\sim$100K parameters. (c) Data efficiency: 1% vs. 10% training data at growth rate 12. (d) Ranking at the $\sim$100K parameter budget.

5.2 3D organ classification under octahedral symmetry

The PatchCamelyon experiment validates bispectral pooling for 2D cyclic/dihedral groups. To evaluate the main technical contribution of the library—the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$)—we design a controlled 3D experiment on OrganMNIST3D **?**, a benchmark of 1,742 abdominal CT organ volumes (28^3 voxels, 11 classes; train/val/test = 971/161/610). Rotation invariance is physically meaningful here: the same organ may appear at different orientations depending on patient positioning, and a classifier should be insensitive to such transformations.

Architecture and experimental design. All equivariant variants share an O-equivariant 3D ResNet backbone. A *lifting convolution* maps the scalar input to a regular representation over O (24 copies per channel), followed by two stages of group-equivariant residual blocks with 3^3 kernels, equivariant batch normalization, and spatial average pooling between stages. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts. The four model variants differ *only* in the final invariant pooling layer:

- **Standard**: a plain 3D CNN (no equivariance) trained with random octahedral augmentation and global average pooling (16K parameters).
- **Max Pool**: O-equivariant backbone with element-wise max over the group dimension (374K parameters).
- **Norm Pool**: O-equivariant backbone with channel-wise ℓ^2-norm over the group dimension (375K parameters).
- **Bispectrum**: O-equivariant backbone with the selective G-bispectrum via `OctaonOcta`, producing 172 invariant scalars per spatial location (463K parameters).

Table 4: Published baselines on OrganMNIST3D for context. Our models operate at $\leq$463K parameters; accuracy differences with respect to large-scale models reflect this deliberate constraint.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov 2024	38K	0.879	0.992
EquiLoPO (local train.)	ICLR 2025	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. 2023	—	0.745	—
Regular SE(3) conv	Kuipers & Bekkers 2023	172K	0.698	—

Table 5: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot} are the mean and standard deviation of accuracy across all 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005

All models are trained with AdamW (lr=10^{-3}, weight decay 10^{-4}) for up to 100 epochs with early stopping on validation AUC (patience 15). Each configuration is run with three seeds (42, 123, 456). At test time, we evaluate rotation robustness by applying all 24 octahedral rotations to the test set and reporting the mean accuracy and its standard deviation across rotations.

Published baselines. Table 4 situates our results against published methods on the same benchmark. Note that large models such as ResNet-18+3D (33M parameters) achieve $\sim$90% accuracy; our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale: Regular SE(3) convolutions **?** reach 69.8% with 172K parameters.

Results. Table 5 reports our ablation results averaged over three seeds. The key findings are:

Rotation invariance. All three equivariant models achieve *exact* rotation invariance: $\sigma_{\mathrm{rot}} = 0.0000$ for max pool and norm pool, and $\sigma_{\mathrm{rot}} = 0.0005$ for the bispectrum (residual floating-point variation). The standard CNN, despite octahedral augmentation during training, drops $\sim$2.5 percentage points under rotation with $\sigma_{\mathrm{rot}} = 0.0115$—augmentation helps but does not guarantee invariance.

Accuracy and completeness. The bispectrum model (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%). Norm pool also exhibits high seed variance ($\pm$9.9%, with one seed at 42.8%), suggesting it is a less stable invariant map. The near-parity of bispectrum and max pool is notable: the bispectrum provides *complete* invariants (no equivariant information is discarded), whereas max pooling is inherently lossy. In settings where downstream tasks depend on fine-grained orientation features—which this 11-class organ classification may not fully stress—completeness could provide a decisive advantage.

Parameter efficiency. At 463K parameters, the bispectrum model outperforms Regular SE(3) convolutions (69.8% at 172K) and approaches SE3MovFrNet (74.5%), validating `OctaonOcta` as a practical invariant pooling primitive for 3D octahedral equivariance.

Data efficiency: when completeness matters. The full-data results show bispectrum and max pool at near-parity, raising the question of whether completeness provides any practical benefit. To stress this, we vary the training set fraction while keeping the architecture and hyperparameters fixed (channels $(4, 8)$, seed 42). Table 6 and Figure 3 report test accuracy across fractions.

The pattern is clear and consistent across seeds: at 5% of training data (48 samples), the bispectrum reaches $33.2 \pm 2.2\%$ while max pool achieves only $18.9 \pm 7.2\%$. At 10% the gap grows to $49.7 \pm 4.4\%$ vs. $42.0 \pm 8.1\%$, and at 25% the bispectrum leads $57.3 \pm 4.7\%$ vs. $45.1 \pm 12.6\%$. Only at full data does max pool catch up ($73.0 \pm 3.3\%$ vs. $72.6 \pm 2.7\%$). Notably, bispectrum's standard deviations are consistently smaller, indicating more stable training in the low-data regime.

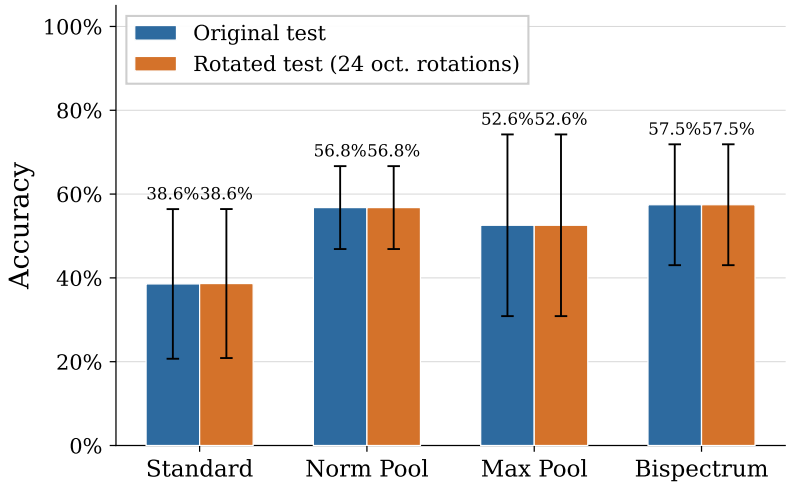

Figure 2: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

Table 6: Data efficiency on OrganMNIST3D (channels $(4, 8)$, mean $\pm$ std over 3 seeds). Bispectrum's advantage is largest in the low-data regime, where its complete invariant map extracts more information per sample.

Fraction	Samples	Standard	Max Pool	Bispectrum
5%	48	$.186 \pm .096$	$.189 \pm .072$	$\mathbf{.332 \pm .022}$
10%	97	$.213 \pm .111$	$.420 \pm .081$	$\mathbf{.497 \pm .044}$
25%	242	$.427 \pm .077$	$.451 \pm .126$	$\mathbf{.573 \pm .047}$
50%	485	$.502 \pm .020$	$.662 \pm .016$	$\mathbf{.639 \pm .057}$
100%	971	$.601 \pm .017$	$\mathbf{.730 \pm .033}$	$.726 \pm .027$

This behavior directly reflects the theoretical distinction between the two pooling strategies. Max pooling is a *lossy* invariant: it discards all group-equivariant structure except the channel-wise maximum and must compensate by learning to reconstruct relevant features from data. When data is abundant, this compensation succeeds. The bispectrum, by contrast, is a *complete* invariant: it preserves all orientation information as polynomial invariants, requiring no learned compensation. In the low-data regime, this prior knowledge about group structure substitutes for missing training examples, yielding substantially better sample efficiency.

Wider channels. To test whether the advantage of bispectral completeness persists at higher capacity, we also run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data. At $(8, 16)$ the two methods are equivalent (75.3% bispectrum vs. 74.9% max pool, both at $\sim$1.5M parameters). At $(16, 32)$ max pool reaches 83.0% while bispectrum drops to 71.5%—the 6.3M-parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings above.

6 Related Work

TODO: Write this section around software/tools and practical distinctions, not around the abandoned steerable-to-Fourier narrative.

Suggested structure:

- equivariant deep learning libraries such as `e2cnn`/`escnn`,
- classical and modern bispectrum theory,
- selective bispectrum papers for finite groups and the disk,

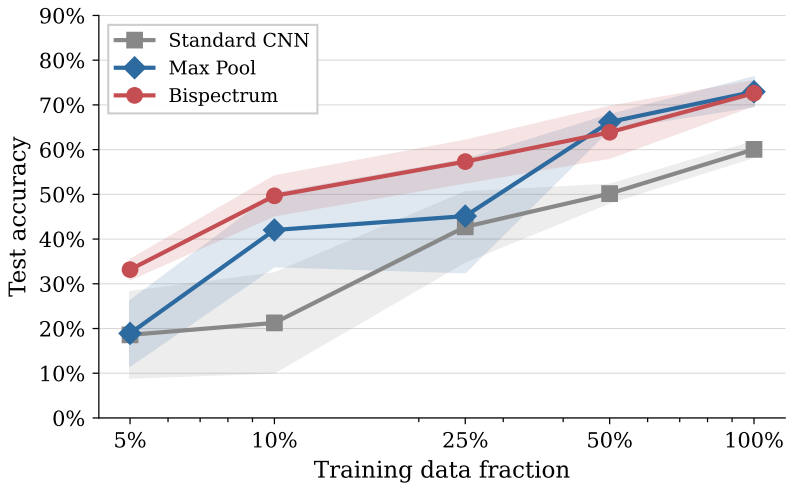

Figure 3: Test accuracy vs. training data fraction on OrganMNIST3D (mean over 3 seeds; shaded bands show ± 1 std). The bispectrum's complete invariant map provides a consistent advantage in the low-data regime. The gap narrows as more data becomes available and max pool can learn to approximate the relevant invariants from examples alone.

- bispectral CNNs in the Depeursinge/Oreiller line,
- tensor-product and Clebsch–Gordan nonlinearities,
- software-paper precedents at NeurIPS.

Nina: How aggressively should we distinguish this paper from prior bispectral CNN work?

Nina: Should we explicitly say that our contribution is breadth of group/domain support + inversion + PyTorch integration, rather than claiming the general idea of bispectral invariants in CNNs is new?

7 Discussion and Limitations

TODO: Write this section early, not last. It can help constrain the rest of the paper.

Minimum limitations to acknowledge:

- selective $\mathrm{SO}(3)$ on S^2 remains open,
- inversion is not available for every supported transform,
- empirical evaluation is currently narrower than the full intended application scope of the library,
- some benchmark or experimental claims may need to be trimmed to match the completed evidence.

Nina: Should the limitations section explicitly state that the paper does not attempt a general theory of bispectral pooling for arbitrary steerable representations?

Roadmap.

TODO: Add a brief roadmap paragraph for PI/co-author review.

Candidate roadmap items:

- finish stable benchmark scripts and plots,
- finish matched PCam sweeps,
- improve docs/examples for public release,
- future work on selective $\mathrm{SO}(3)$ and richer empirical applications.

8 Conclusion

TODO: Write a short conclusion once the final scope is fixed.

Proposed conclusion shape:

- restate the library contribution,
- emphasize octahedral inversion as the main technical case study,
- summarize what the completed experiments support,
- end with realistic next steps rather than broad claims.

References

Bastiaan S. Veeling, Jasper Linmans, Jim Winkens, Taco Cohen, and Max Welling. Rotation equivariant CNNs for digital pathology. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 210–218. Springer, 2018.

A Implementation Details

TODO: Collect package-level implementation details that do not fit in the main paper: buffers, transforms, inversion APIs, optional dependencies, dtype/device behavior.

B Benchmark Details

TODO: Include full benchmark tables only after the benchmark script is cleaned up and the figures are final.

C Experimental Details

TODO: Include architecture details, hyperparameters, seeds, train fractions, hardware, and exact result aggregation procedures.

Nina: Do we want appendix-level tables for every completed sweep, or only the ones directly supporting the paper's final claims?

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

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   Answer: **[TODO]**

Justification: **[TODO]**

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- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
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